Mohammad Ful Hossain Seikh

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ORCID INSPIRE-HEP Google Scholar ResearchGate Scopus

Education

Ph.D. in Physics *(2020–2026)*
University of Kansas, USA | GPA: 3.964/4.0

M.S. in Electrical Engineering *(2024–2025)*
University of Kansas, USA | GPA: 3.778/4.0

M.Sc. in Physics *(2018–2020)*
Aligarh Muslim University, India | GPA: 9.170/10 *(Gold Medalist)*

B.Sc. Honors in Physics *(2015–2018)*
Aligarh Muslim University, India | GPA: 9.186/10 *(Gold Medalist)*

Research Experience

Research Roles and Appointments

- **Postdoctoral Researcher: Particle Astrophysics, University of Kansas** *(Jun 2026–Present)*
  Study radio detection of ultra-high energy particles, backgrounds, and detector response with Prof. D. Besson.
- **Core Analyst, ARA Five-Station Diffuse Neutrino Search** *(2022–Present)*
  One of eight core analysts for the 10.6-year five-station ARA diffuse search. Integrated ARA01 into the common framework and contributed to optimization, unblinding, and interpretation.
- **Analysis Lead, Non-Anthropogenic Down-going Signals in ARA** *(Jul 2026–Present)*
  Leading an array-wide search across all five ARA stations for down-going radio signals inconsistent with known anthropogenic backgrounds.
- **Analysis Lead, Radio Background Characterization with ARA** *(Dec 2025–Present)*
  Leading studies of astrophysical, environmental, and anthropogenic radio backgrounds in ARA data, including solar activity, galactic noise, high wind triboelectric events, radiosondes, and local sources.
- **Primary Analyzer, Askaryan Radio Array (ARA) Station 1** *(2021–2026)*
  Led ARA01 calibration, data-driven detector modeling, reconstruction, background rejection, and simulation. Developed its analysis framework for the array-wide search.
- **Monitoring Shifter, IceCube Neutrino Observatory** *(2021–2026)*
  Served as KU representative for monitoring shifts, presenting weekly reports to the collaboration.
- **Visiting Summer Student** *(May–Jul 2019)*
  Studied "Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm" with Prof. S.K. Agarwalla at the Institute of Physics, India, producing technical report `DOI:10.13140/RG.2.2.34421.12009`.
- **Winter Research Intern** *(Dec 2019–Jan 2020)*
  Researched "Weak Interaction Physics" at Indian Institute of Technology–BHU with Prof. G. Abbas.

Scientific Software and Open-Source Development

- **MAHAM: Modules for Astroparticle, High-energy, Astronomy and Multimessenger** *(2026–Present)*
  Founder and lead developer of `Project MAHAM`, an open-source framework for validated and reusable methods in astroparticle physics and multimessenger astronomy. Designed its architecture and provenance checks; developing neutrino spectra, experimental limits, detector response, statistics, and literature datasets.
- **AAFIYA: Antenna Analysis in Frequency-domain for Impedance & Yield Assessment** *(2024–Present)*
  Creator of `AAFIYA`, an open-source toolkit for antenna measurements and simulations, including S-parameters, impedance, gain patterns, polarization, vector effective length, and measurement-to-simulation comparisons.
- **AraProc: ARA Data Analysis Framework** *(2024–2026)*
  Co-developer of the ARA Collaboration's `AraProc` Python analysis framework, with 21 merged pull requests covering waveform analysis, event reconstruction, CW filtering, and data quality improvements.

Field Experience

- **NSF's Summit Station, Greenland Field Campaign** *(Apr–May 2025)*
  Participated in the RNO-G calibration campaign, conducting in-situ radio frequency measurements, setting up measurement instruments, and deploying equipment at Summit Station, Greenland.

Selected Dissertations, Theses, and Projects

- **Physics PhD Dissertation** *(2020–2026)*
  "Toward a Five-Station Diffuse Ultra-High Energy Neutrino Search with the Askaryan Radio Array: ARA01 Analysis Framework." Defended May 8, 2026. Advised by Prof. D. Besson. Published on ProQuest ISBN: 9798252480190 and KU ScholarWorks.
- **Electrical Engineering Master's Project** *(2024–2025)*
  Developed **AAFIYA** for antenna measurement and simulation with Prof. Jim Stiles (KU EECS), leading to a paper published in the IEEE AP-S/URSI 2026 proceedings. (GitHub), (Report), (Slides).
- **Machine Learning Course Project** *(2025)*
  Developed a CNN model for "Automatic Surface Detection in Echograms using Deep Learning" in a KU EECS machine learning course. (GitHub), (Report).
- **Physics Master's Thesis** *(2019–2020)*
  Investigated light sterile neutrino oscillations with Prof. M.S. Athar at Aligarh Muslim University (AMU), producing review report "Light Sterile Neutrinos: Status and Prospects." DOI:10.13140/RG.2.2.13777.66405.
- **Physics Undergraduate Project** *(2017–2018)*
  "Quarks & Leptons": Standard Model study with Prof. M.S. Athar, AMU (Handwritten Report).

Publications & Manuscripts

First author works and core analysis manuscripts, followed by selected collaborative publications.

Manuscripts in Collaboration Review

Core analysis team: Bishop, A., Clark, B.A., Dasgupta, P., Giri, P., Machtay, A., Muzio, M.S., Salcedo-Gomez, A., and **Seikh, M.F.H.**

- **ARA Collaboration** *(2026)*. "First Array-Wide Search for Ultrahigh Energy Neutrinos with the Askaryan Radio Array." *In collaboration review; planned submission to Physical Review Letters.*
- **ARA Collaboration** *(2026)*. "Analysis Methodology of the Search for Ultrahigh Energy Neutrinos in 10.6 Years of Askaryan Radio Array Data." *In collaboration review; planned submission to Physical Review D.*

Refereed Journal Publications (First Author)

[3] **Seikh, M.F.H.** *(2025)*. "A geometrical approach to neutrino oscillation parameters." *AIP Advances 15(10), 105204*. DOI:10.1063/5.0292233; arXiv:2510.06526.

[2] **Seikh, M.F.H.** *(2025)*. "Physics of radio antennas." *Eur. Phys. J. Spec. Top. 234(16), 4967–4983*. DOI:10.1140/epjs/s11734-025-01500-7; arXiv:2411.07507.

[1] **Seikh, M.F.H.** for the ARA Collaboration *(2025)*. "Askaryan Radio Array: searching for the highest energy neutrinos." *Eur. Phys. J. Spec. Top. 234(16), 5065–5075*. DOI:10.1140/epjs/s11734-025-01708-7; arXiv:2411.01761.

Conference Proceedings (First/Lead Author)

[4] **Seikh, M.F.H.**, Jarvis, R., & Stiles, J. *(2026)*. "AAFIYA: Antenna Analysis in Frequency-Domain for Impedance and Yield Assessment." *IEEE AP-S/URSI*, pp. 2537–2540. DOI:10.1109/AP-S/USNC-URSI60190.2026.11675620; arXiv:2601.13583.

[3] **Seikh, M.F.H.**, Giri, P., & Besson, D. for the ARA Collaboration *(2025)*. "Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array." PoS(ICRC2025)1171; arXiv:2509.20735.

[2] **Seikh, M.F.H.** & Besson, D. for the RNO-G Collaboration *(2025)*. "Observation of Radio Solar Flares in RNO-G." PoS(ICRC2025)1365.

[1] **Seikh, M.F.H.** for the ARA Collaboration *(2023)*. "Calibration and Physics with ARA Station 1." PoS(ICRC2023)1163; arXiv:2308.07292.

Editorial and Other Scientific Writing

[5] **Seikh, M.F.H.** & Besson, D. *(2026).* "Polar Ice as a Laboratory for Physics." *Physics Bulletin 2026, AMU*, pp. 27–37.

[4] **Seikh, M.F.H.**, Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D. *(2025).* "First Radio Neutrino & Cosmic Rays Astronomy Workshop in India." *Eur. Phys. J. Spec. Top. 234(16), 4919–4924.* DOI:10.1140/epjs/s11734-025-01955-8; arXiv:2509.20704. *(Editorial for the special issue.)*

[3] **Seikh, M.F.H.** *(2025).* "From Aligarh to the Arctic: Carrying the Spirit of Sir Syed to Greenland's Summit." *ALIG Atlanta 2025*, pp. 20–21.

[2] **Seikh, M.F.H.** & Besson, D. *(2024).* "IceCube Neutrino Observatory: A Cold Discovery Hub." *Physics Bulletin 2024, AMU*, pp. 14–23.

[1] **Seikh, M.F.H.** *(2020).* "Neutrinos and the 40 Future Routes." *Physics Bulletin, AMU.* DOI:10.13140/RG.2.2.27981.10729.

Selected Collaborative Publications

Collaboration authored and multi-institution papers; these are distinct from the first author works above.

[11] **ARA Collaboration** *(2026).* "Observation of In-Ice Askaryan Radiation from High-Energy Cosmic Rays." *Phys. Rev. Lett. 136, 151003.* DOI:10.1103/xwqy-yzrk; arXiv:2510.21104.

[10] **RNO-G Collaboration** *(2026).* "Probing the firn refractive index profile using antenna response." *J. Glaciol. 72, e39.* DOI:10.1017/jog.2026.10133; arXiv:2504.03862.

[9] Hornhuber, C., **Seikh, M.F.H.**, Stockham, M., Voigt, S., Young, R., et al. *(2026).* "Development of a custom kV-amplitude, pressure-tolerant radio-frequency transmitter." *Nucl. Instrum. Meth. A 1089, 171584.* DOI:10.1016/j.nima.2026.171584; arXiv:2512.14866.

[8] **RNO-G Collaboration** *(2025).* "Instrument design and performance of the first seven stations of RNO-G." *JINST 20(04), P04015.* DOI:10.1088/1748-0221/20/04/P04015; arXiv:2411.12922.

[7] **RNO-G Collaboration** *(2025).* "Solar flare observations with the Radio Neutrino Observatory Greenland (RNO-G)." *Astropart. Phys. 164, 103024.* DOI:10.1016/j.astropartphys.2024.103024; arXiv:2404.14995.

[6] **IceCube Collaboration** *(2025).* "Search for Extremely-High-Energy Neutrinos and First Constraints on the Ultrahigh-Energy Cosmic-Ray Proton Fraction with IceCube." *Phys. Rev. Lett. 135, 031001.* DOI:10.1103/PhysRevLett.135.031001; arXiv:2502.01963.

[5] **IceCube-Gen2 Collaboration** *(2025).* "Probing ultra-high-energy neutrinos with the IceCube-Gen2 in-ice radio array." PoS(ICRC2025)1045; arXiv:2507.07813.

[4] **PUEO Collaboration** *(2025).* "The Payload for Ultrahigh Energy Observations: Detector Design and Implementation." PoS(ICRC2025)1063.

[3] **RET Collaboration** *(2024).* "Macroscopic approach to the radar echo scatter from high-energy particle cascades." *Phys. Rev. D 109, 083012.* DOI:10.1103/PhysRevD.109.083012; arXiv:2310.06731.

[2] **ARA Collaboration** *(2022).* "Low-threshold ultrahigh-energy neutrino search with the Askaryan Radio Array." *Phys. Rev. D 105, 122006.* DOI:10.1103/PhysRevD.105.122006; arXiv:2202.07080.

[1] **TAROGE-M/ARIANNA Collaborations** *(2022).* "TAROGE-M: radio antenna array on antarctic high mountain for detecting near-horizontal ultra-high energy air showers." *JCAP 11, 022.* DOI:10.1088/1475-7516/2022/11/022; arXiv:2207.10616.

The complete publication record, including contributions to the ARA, RNO-G, ARIANNA, IceCube, IceCube-Gen2, PUEO, TAROGE-M, and RET Collaborations, is available in the Complete Publication List (PDF).

Invited Talks, Seminars, & Lectures

Upcoming Invitations

- **Neutrino Searches and Radio Backgrounds in the Askaryan Radio Array**
  HEP Seminar, Oklahoma State University *(Nov 5, 2026)* | *scheduled*
- **Neutrinos in Multimessenger Astronomy**
  Guest Lectures, ASTR 798 High-Energy Astrophysics, University of Kansas *(Dec 1 & 3, 2026)* | *scheduled*
- **WUCAP Seminar**, Washington University in St. Louis *(Feb 12, 2027)* | *scheduled*
- **Physics Colloquium**, Missouri University of Science and Technology *(Spring 2027)* | *date TBD*

Delivered Invited Talks

[9] **The Search for the Highest-Energy Neutrinos in Polar Ice**
Eddy and April Lucas Physics Seminar, Wichita State University *(Oct 21, 2026)*

[8] **Searching for the Universe's Highest-Energy Neutrinos with Radio Detectors**
Particle Physics Seminar, Carnegie Mellon University *(Sep 9, 2026)*

[7] **Radio Engineering for Particle Astrophysics Experiments**
HEP Seminar, University of Nebraska-Lincoln *(Mar 26, 2025)*

[6] **Radio Engineering for Particle Astrophysics Experiments**
Astroparticle Seminar, Ohio State University *(Jan 22, 2025)*

[5] **Radio Engineering & Antenna Physics**
Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage *(Jul 2–5, 2024)*

[4] **UHE Neutrino & Cosmic Ray Detectors**
ICEHAP Seminar, Chiba University, Japan *(Aug 9, 2023)*

[3] **Antenna Physics & Measurements**
Lecture Series, IEI Summer School 2023, South Dakota School of Mines *(Jul 2–5, 2023)*

[2] **UHE Neutrino & Cosmic Ray Detectors**
Physics Colloquium, AMU, India *(Jan 12, 2023)*

[1] **Antennas in UHE Neutrino Detectors**
Particle Physics Seminar, University of Delaware *(Oct 4, 2022)*

Teaching & Research Mentoring

Teaching & Tutoring

- **PHSX 211: General Physics I, KU** *(Fall 2021)*
  Course Instructor and Discussion Moderator. Classical mechanics and thermodynamics for engineering and physical science majors.
- **PHSX 115: College Physics II, KU** *(Fall 2021)*
  Lab Instructor. Electricity, magnetism, optics, and introductory atomic and nuclear physics.
- **PHSX 114: College Physics I, KU** *(Summer 2021)*
  Lab Instructor. Mechanics, fluids, heat, thermodynamics, and sound.
- **Private Tutor, Aligarh, India** *(2016–2020)*
  Taught high school physics and mathematics during B.Sc. and M.Sc. studies.

Research Advising

- **Daniel Dakroub**, Senior Undergraduate, MEPhI, Moscow, Russia *(Apr 2026–Present)*
  Project: Inter-Experiment Search for Surface-Mode Radio Propagation.
- **Burhan-Ullah Khan**, High School Student, RMPS AMU City School, Aligarh, India *(Sep 2026–Present)*
  Project: Model Development for Explaining ANITA Anomalous Events.

Professional Service

Editorial

- **Special-Issue Co-editor** of *The European Physical Journal Special Topics*: "Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray" (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Kravchenko (View Cover) *(2024–2025)*
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU's Society of Physics Students (View Issue) *(Jun 2022)*

Peer Review

- **Peer Reviewer**, *International Journal of Modern Physics A* (1 manuscript, 2 rounds) *(2025)*
- **Peer Reviewer**, *The European Physical Journal Special Topics* (5 manuscripts, 11 rounds) *(2024–2025)*

Internal Collaboration Review

- Muzio, M.S. and Salcedo-Gomez, A. for the ARA Collaboration, "The First Array-Wide Diffuse Flux Search for UHE Neutrinos with the Askaryan Radio Array", ARENA *(Jun 2026)*
- Giri, P. for the ARA Collaboration, "A Comprehensive Diffuse Neutrino Search Using the Full Askaryan Radio Array", Lepton–Photon *(Nov 2025)*

Conference Session Chairing

- **General Cosmology:** for APS Global Physics Summit *(Mar 19, 2026)*
- **Topical Groups Session II:** for APS Global Physics Summit *(Mar 17, 2026)*

Application Review

- **REU Application Reviewer:** for KU Physics and Astronomy (10 Applications) *(2025)*
- **SSGSA Application Reviewer:** for SSGSA (70+ Applications) *(2021–2026)*

Professional Leadership & Membership

- **Workshop Co-organizer and Host**: India's first UHE particle radio-detection workshop, AMU*(Apr 2024)*.
- **American Physical Society (APS)**: Member *(2020–Present)*
- **National Society of Black Physicists**: Member *(2021–Present)*
- **Physics & Astronomy Graduate Student Organization (GSO), KU**:
  - ✓ Graduate Students Representative - Member at Large *(2025–2026)*
  - ✓ Graduate Students Representative - Liaison *(2024–2025)*
  - ✓ GSO Executive Officer *(2023–2024)*
- **Sir Syed Global Scholar Award (SSGSA)**:
  - ✓ Peer Mentor *(2023–Present)*
  - ✓ Team Lead, Promotions & Outreach *(2023–2025)*
- **Society of Physics Students, KU Chapter**: Graduate President *(2021–2022)*
- **Sigma Pi Sigma (National Physics Honor Society)**: Lifetime Member
- **AMU Society of Physics**: Member *(2018–Present)*

Technical Proficiencies

- **Programming**: Python, C/C++, Fortran
- **Scientific Computing & Data Analysis**: CERN ROOT, uproot, NumPy, SciPy, pandas, scikit-learn
- **Detector, Simulation & RF Software**: AraRoot/AraProc, AraSim, NuRadioMC, HFSS, Mathematica
- **Development & Computing**: Git/GitHub, Linux (Ubuntu), macOS, Windows
- **Document Preparation**: LaTeX, Microsoft Office Suite

Conference Presentations

Oral Presentations

[11] **Pioneering an Array-Wide Search for UHEν with ARA**. APS Global Physics Summit *(2026)*.
[10] **RNO-G in-ice Antennas**. RNO-G Collaboration Meeting at Uppsala *(2024)*.
[9] **Radio Frequency Instrumentation**. Radio Neutrino Workshop at AMU *(2024)*.
[8] **Towards Five Station ARA Analysis**. APS April Meeting *(2024)*.
[7] **ARA Antenna Models**. ARA Collaboration Meeting at Ohio State *(2024)*.
[6] **Current Status & Future of ARA**. APS Prairie Section Meeting at U Missouri *(2023)*.
[5] **Introduction to Antenna Physics**. RET Collaboration Meeting at VUB *(2023)*.
[4] **Gain Calibration of RNO-G Antennas**. RNO-G Fall Meeting at Penn State *(2023)*.
[3] **Galactic Noise in Radio Neutrino Experiments**. Astro Seminar at KU *(2022)*.
[2] **Towards ARA Station 1 Calibration**. APS April Meeting *(2022)*.
[1] **RNO-G Ice Calibration Planning**. RNO-G Collaboration Meeting at U Chicago *(2022)*.

Poster Presentations

[8] **A Diffuse Search for UHE Cosmogenic Neutrinos with ARA Five Stations**. Neutrino *(2026)*.
[7] **Monitoring Snow Accumulation in ARA Using DAQ Temperature Data**. Neutrino *(2026)*.
[6] **A Geometrical Approach to Neutrino Oscillation Parameters**. APS Global Physics Summit *(2026)*.
[5] **CW Identification & Filtering Techniques in ARA**. 39th ICRC *(2025)*.
[4] **Observation of radio solar flares in RNO-G**. 39th ICRC *(2025)*.
[3] **Gain Measurements of LPDA Antennas**. APS April Meeting *(2024)*.
[2] **Overview of ARA Detector: Five Station Analysis**. ICTS Neutrino School *(2024)*.
[1] **Calibration of ARA Station 1**. 38th ICRC *(2023)*.

Honors & Awards

Academic Honors

- **University Gold Medals**:
  - ✓ M.Sc. Physics (Top among 50 students), AMU *(2020)*
  - ✓ B.Sc. Honors Physics (Top among 120 students), AMU *(2018)*
  - ✓ Highest percentage in B.Sc. (Hons) Faculty of Science, AMU *(2018)*
- **Abdul Aziz Medal** for B.Sc. Physics, AMU *(2018)*
- **Undergraduate Topper Scholarship**, UGC India *(2020)*

Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics *(2025–2026)*
- **McKeithan Summer Fellowship**, KU Physics *(2025)*
- **Graduate Student Award for Distinguished Service**, KU *(2024–2025)*
- **Wallace S. Strobel Scholarship**, KU EECS *(2024–2025)*
- **Ralston Summer Fellowship**, KU Physics *(2024)*
- **Doctoral Student Research Fund**, KU *(2024)*
- **Distinguished Students Award**, APS FIP *(2021–2022)*
- **Carl & Catherine Chaffee Scholarship**, KU Physics *(2020–2021)*
- **University Merit Financial Award**, AMU *(2019–2020)*

Conference Travel Awards

- **Junior Participant Support**, Neutrino *(2026)*
- **Graduate Student Travel Fund**, KU Physics *(2023, 2025)*
- **DAP Travel Grant**, APS *(2024)*
- **DPF Travel Award**, APS *(2022, 2024)*
- **CLAS Graduate Scholarly Development Fund**, KU *(2022)*

Other Distinctions

- **Washington DC Scholarship**, AMU Alumni Association *(2015–2019)*
- **Sir Syed Global Scholar Award** for prospective graduate students *(2018)*
- **Senior Diploma in Painting with Distinction**, Bangiya Sangeet Parishad *(2014)*